The Transportation Algorithm

- The dual problem

Consider the transportation problem and its dual:

$$\min \ z(x) = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij} \qquad\qquad \max \ z(u,v) = \sum_{i=1}^{m} a_i u_i + \sum_{j=1}^{n} b_j v_j \quad (D)$$

$$\text{s.t.} \ \sum_{j=1}^{n} x_{ij} = a_i, \quad i=1,\cdots,m \qquad u_i \qquad\qquad \text{s.t.} \ u_i + v_j \le c_{ij} \ \text{for all } i,j$$

$$(P)$$

$$\sum_{i=1}^{m} x_{ij} = b_j, \quad j=1,\cdots,n \qquad v_j$$

$$x_{ij} \ge 0$$

The complementary slackness conditions are:

$$x_{ij}(c_{ij} - u_i - v_j) = 0 \quad \text{for all } i,j$$

- Finding a dual solution

- Recall that the simplex method keeps primal feasibility and complementary slackness, and works towards dual feasibility. For each basic feasible solution $\bar{x}$, there is a corresponding dual solution $(\bar{u},\bar{v})$ such that $(\bar{x},\bar{u},\bar{v})$ satisfies the complementary slackness conditions. Since $\bar{x}_{ij}=0$ for nonbasic variables, it is sufficient for $(\bar{u},\bar{v})$ to satisfy

$$c_{ij} - u_i - v_j = 0$$

for all (i,j) such that x_{ij} is a basic variable.

Note that for a BFS, there are $m+n-1$ basic variables. Therefore, there are $m+n-1$ equations about the $m+n$ dual variables u_i $(i=1,\cdots,m)$ and v_j $(j=1,\cdots,n)$. We can assign an arbitrary value to any one of the dual variables (e.g.: $u_1=0$) and solve for the others.

e.g. Consider a transportation problem with a BFS in the following transportation tableau.

	10	30	20
20	$\boxed{2}$ 10	$\boxed{4}$ 10	$\boxed{3}$
20	$\boxed{1}$	$\boxed{5}$ 20	$\boxed{2}$ 0
20	$\boxed{1}$	$\boxed{1}$	$\boxed{6}$ 20

We can calculate a dual solution by a chain of simple operations.

$$u_1 = 0$$

$$2 = c_{11} = u_1 + v_1 \Rightarrow v_1 = 2$$

$$4 = c_{12} = u_1 + v_2 \Rightarrow v_2 = 4$$

$$5 = c_{22} = u_2 + v_2 \Rightarrow u_2 = 1$$

$$2 = c_{23} = u_2 + v_3 \Rightarrow v_3 = 1$$

$$6 = c_{33} = u_3 + v_3 \Rightarrow u_3 = 5$$

We mark the dual solution around the transportation tableau.

	10	30	20		
20	$\boxed{2}$ 10	$\boxed{4}$ 10	$\boxed{3}$	0	u_1
20	$\boxed{1}$	$\boxed{5}$ 20	$\boxed{2}$ 0	1	u_2
20	$\boxed{1}$	$\boxed{1}$	$\boxed{6}$ 20	5	u_3
	2	4	1		
	v_1	v_2	v_3		

- Checking for dual feasibility

For each primal variable x_{ij}, the corresponding dual constraint is $c_{ij} - u_i - v_j \geq 0$.

If $\overline{c_{ij}} := c_{ij} - u_i - v_j \geq 0$ for all nonbasic variables, then the current solution is optimal.

e.g.: Consider the previous example.

$$\overline{c_{13}} = c_{13} - u_1 - v_3 = 3 - 0 - 1 = 2$$

$$\overline{c_{21}} = c_{21} - u_2 - v_1 = 1 - 1 - 2 = -2$$

$$\overline{c_{31}} = c_{31} - u_3 - v_1 = 1 - 5 - 2 = -6$$

$$\overline{c_{32}} = c_{32} - u_3 - v_2 = 1 - 5 - 4 = -8$$

We put $\overline{c_{ij}}$ on the upper right corner of cell (i,j)

	10	30	20		
20	$\boxed{2}$ 10	$\boxed{4}$ 10	$\boxed{3}$ $\boxed{2}$	0	u_1
20	$\boxed{1}$ $\boxed{-2}$	$\boxed{5}$ 20	$\boxed{2}$ 0	1	u_2
20	$\boxed{1}$ $\boxed{-6}$	$\boxed{1}$ $\boxed{-8}$	$\boxed{6}$ 20	5	u_3
	2	4	1		
	v_1	v_2	v_3		

- Transportation algorithm

Step 0 Find an initial basic feasible solution x

(northwest corner rule)

	10	30	20
20	$\boxed{2}$ 10	$\boxed{4}$ 10	$\boxed{3}$
20	$\boxed{1}$	$\boxed{5}$ 20	$\boxed{2}$ 0
20	$\boxed{1}$	$\boxed{1}$	$\boxed{6}$ 20

Step 1 Determine the dual solution

Find (u, v) such that

$$c_{ij} - u_i - v_j = 0$$

for each basic cell (i,j).

	10	30	20		
20	$\boxed{2}$ 10	$\boxed{4}$ 10	$\boxed{3}$	0	u_1
20	$\boxed{1}$	$\boxed{5}$ 20	$\boxed{2}$ 0	1	u_2
20	$\boxed{1}$	$\boxed{1}$	$\boxed{6}$ 20	5	u_3
	2	4	1		
	v_1	v_2	v_3		

Step 2 Check for optimality

Compute $\overline{c_{ij}} = c_{ij} - u_i - v_j$ for each nonbasic cell (i,j)

If $\overline{c_{ij}} \geq 0$ for all (i,j),

then (u,v) is feasible for (D)

and x is optimal for (P) STOP

	10	30	20	
20	$\boxed{2}$ 10	$\boxed{4}$ 10	$\boxed{3}$ $\boxed{2}$	0
20	$\boxed{1}$ $\boxed{-2}$	$\boxed{5}$ 20	$\boxed{2}$ 0	1
20	$\boxed{1}$ $\boxed{-6}$	$\boxed{1}$ $\boxed{-8}$	$\boxed{6}$ 20	5
	2	4	1	

Step 3 Determine the entering variable

Choose x_{ij} with $\overline{c_{ij}} < 0$

as the entering variable

We usually pick the one with

the most negative $\overline{c_{ij}}$

	10	30	20	
20	$\boxed{2}$ 10	$\boxed{4}$ 10	$\boxed{3}$ $\boxed{2}$	0
20	$\boxed{1}$ $\boxed{-2}$	$\boxed{5}$ 20	$\boxed{2}$ 0	1
20	$\boxed{1}$ $\boxed{-6}$	$\boxed{1}$ $\boxed{-8}$	$\boxed{6}$ 20	5
	2	4	1	

Step 4 Determine the departing variable

- Find the unique loop created by the entering cell and some other basic cells.
- Add Δ units to the entering cell,

subtract Δ units from the next cell in the loop,

add Δ units to the next, subtract Δ units from the next -

until every cell in the loop is visited

	10	30	20	
20	$\boxed{2}$ 10	$\boxed{4}$ 10	$\boxed{3}$ $\boxed{2}$	0
20	$\boxed{1}$ $\boxed{-2}$	$\boxed{5}$ $20-\Delta$	$\boxed{2}$ $0+\Delta$	1
20	$\boxed{1}$ $\boxed{-6}$	$\boxed{1}$ $\boxed{-8}$ Δ	$\boxed{6}$ $20-\Delta$	5
	2	4	1	

- Find the largest Δ such that all cells stay nonnegative. Update the loop

$$\Delta = \min\{x_{ij} \quad (i,j) \text{ is a decreasing cell}\}$$

$\Delta = 20$

	10	30	20	
20	[2] 10	[4] 10	[3] [2]	0
20	[1] [-2]	[5] 0	[2] 20	1
20	[1] [-6]	[1] [-8] 20	[6] 0	5
	2	4	1	

- Pick a cell (i,j) whose value just drops to 0. (i,j) is the departing cell. Empty the cell

	10	30	20
20	[2] 10	[4] 10	[3]
20	[1]	[5]	[2] 20
20	[1]	[1] 20	[6] 0

Step 5 Return to Step 1

	10	30	20	
20	[2] 10	[4] 10	[3] [-6]	0
20	[1] [6]	[5] [8]	[2] 20	-7
20	[1] [2]	[1] 20	[6] 0	-3
	2	4	9	

$\longrightarrow$

	10	30	20	
20	[2] 10	[4] $10-\Delta$	[3] [-6] Δ	0
20	[1] [6]	[5] [8]	[2] 20	-7
20	[1] [2]	[1] $20+\Delta$	[6] $0-\Delta$	-3
	2	4	9	

$\longrightarrow$

	10	30	20	
20	[2] 10	[4] 10	[3] 0	0
20	[1] [0]	[5] [2]	[2] 20	-1
20	[1] [2]	[1] 20	[6] [3]	-3
	2	4	3	

optimal solution:

$$x_{11}^* = x_{12}^* = 10, \quad x_{23}^* = x_{32}^* = 20$$

$$x_{13}^* = x_{21}^* = x_{22}^* = x_{31}^* = x_{33}^* = 0$$

$$u^* = (0, -1, -3)$$

$$v^* = (2, 4, 3)$$